

FIRM VALUATION, AGAIN

MGMT 648 · October 1

Kerry Back

Thu Oct 1, 2026

VALUING THE WHOLE FIRM

Two claims, one stream

The operations produce one stream of cash. Debt and equity divide it.

Debt

Interest and principal, fixed by contract, paid before anything reaches the shareholders.

Equity

Whatever is left, with no ceiling and no promise.

Value the stream once and then divide it. Forecasting the equity's share directly means forecasting the borrowing and repayment schedule as well.

Leave interest out of the cash flows

Interest leaves the shareholders and arrives at the lenders. Both are inside the group being valued, so the payment nets to zero and never appears.

Operating cash flow less investment

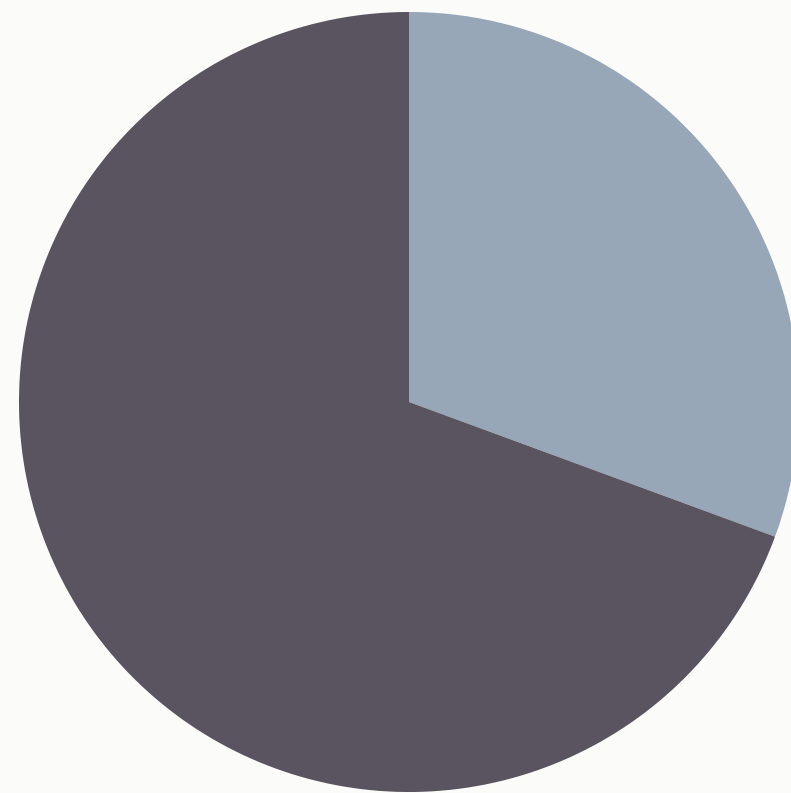


No line for interest, and the tax rate is applied to EBIT rather than to EBIT less interest. The deduction for interest is carried in the discount rate.

The cash flow belongs to both groups, so the rate is the WACC: the blend of what each group requires, weighted by the market value of what each holds. We calculate it on October 10 and take it as given today.

The rate has to match the claim

What the present value buys



ENTERPRISE VALUE

The present value of the free cash flow, discounted at the weighted average rate.

- Debt, at what the lenders are owed
- Equity, the rest

TWO STAGES

Growth above the economy ends

A company growing faster than the economy forever eventually exceeds it. So the forecast is built in two parts, and the second part is the one that has to be sustainable.

- 1 Years 1 to n , forecast each year on its own: sales, margin, the ratios
- 2 Year $n + 1$ onward, one growth rate that never changes

The ceiling on the second stage

2.5%

REAL GDP GROWTH A YEAR

2.2%

THE GDP PRICE DEFLATOR

4.8%

NOMINAL GDP GROWTH

Thirty years through 2025. Real growth and prices compound rather than add:
 $1.025 \times 1.022 = 1.048$. Over the full run since 1947 nominal growth was 6.4% a year, of which 3.1 points were real and 3.2 were prices.

Bureau of Economic Analysis, gross domestic product in current and in chained 2017 dollars, annual. FRED series **GDPA** and **GDPCA**. A second-stage rate above this is a forecast that the company eventually becomes the whole economy.

$$TV_n = \frac{FCF_{n+1}}{r - g}$$

The numerator is the year *after* the last forecast year. The formula assumes *g* is constant forever, that the ratios are constant too, and that *g* stays below *r*, the WACC.

The perpetuity formula

The terminal value is dated n

TV_n is a value at the end of year n , standing in for every cash flow from year $n + 1$ on. It gets discounted n periods, alongside FCF_n .

$$EV = \sum_{t=1}^n \frac{FCF_t}{(1+r)^t} + \frac{TV_n}{(1+r)^n}$$

Discounting the terminal value $n + 1$ periods divides it by an extra $1 + k$. Check the exponent against the exponent on FCF_n : they are the same.

From enterprise value to a price

- 1 Subtract debt, at book value unless it trades far from it
- 2 Add cash and anything else the operations do not use
- 3 Divide by diluted shares

Each balance sheet item belongs in exactly one place: inside the forecast, if it produces the cash flows being discounted, or in this calculation, if it does not. Counting it twice or leaving it out of both are the two failures.

AN EXAMPLE

The assumptions

The forecast

Sales of 100 last year, growing 20%, 20%, 20%, 12%, 8%, and then 4% forever. Operating margin 20%, tax 25%.

The ratios

Net PP&E is 45% of next year's sales. Working capital is 15% of next year's sales.

The rest

WACC 10%. Debt 100, cash 20, ten million shares. Figures in \$ millions.

Every white cell on the slides that follow is editable, and all of them feed one model. Change a number anywhere and the value per share moves everywhere.

Sales and operating profit

LAST YEAR'S SALES 100 GROWTH 20 % 20 % 20 % 12 % 8

% THEN 4 %

OPERATING MARGIN 20 % TAX RATE 25 %

\$ MILLIONS	YR 1	YR 2	YR 3	YR 4	YR 5	YR 6
Sales	120.00	144.00	172.80	193.54	209.02	217.38
EBIT	24.00	28.80	34.56	38.71	41.80	43.48
EBIT after tax (NOPAT)	18.00	21.60	25.92	29.03	31.35	32.61

VALUE PER SHARE **\$25.03**

What the growth costs

NET PP&E, SHARE OF NEXT YEAR'S SALES 45 % WORKING CAPITAL 15 %

\$ MILLIONS	YR 0	YR 1	YR 2	YR 3	YR 4	YR 5	YR 6
Net PP&E	54.00	64.80	77.76	87.09	94.06	97.82	101.73
Working capital	18.00	21.60	25.92	29.03	31.35	32.61	33.91
Invested capital	72.00	86.40	103.68	116.12	125.41	130.43	135.64
Increase	—	14.40	17.28	12.44	9.29	5.02	5.22

VALUE PER SHARE **\$25.03**

Ways to calculate free cash flow

$$\begin{aligned}\text{FCF} &= \text{NOPAT} + \text{Depreciation} - \text{CapEx} - \Delta\text{NWC} \\ &= \text{NOPAT} - \Delta\text{Net PP\&E} - \Delta\text{NWC} \\ &= \text{NOPAT} - \Delta\text{IC}\end{aligned}$$

The first line to the second

Net PP&E rolls forward as last year's balance plus capital expenditure less depreciation. So capital expenditure less depreciation *is* the change in the balance, and two forecast lines collapse into one.

The second line to the third

Net PP&E and net working capital are the two pieces of invested capital, so their changes add to the change in invested capital.

Free cash flow

OPERATING MARGIN 20 % NET PP&E 45 % WORKING CAPITAL 15 %

\$ MILLIONS	YR 1	YR 2	YR 3	YR 4	YR 5	YR 6
EBIT after tax (NOPAT)	18.00	21.60	25.92	29.03	31.35	32.61
Less the increase in invested capital	14.40	17.28	12.44	9.29	5.02	5.22
Free cash flow	3.60	4.32	13.48	19.74	26.34	27.39

VALUE PER SHARE **\$25.03**

Years 1 and 2 grow at 20% and hand back almost nothing. Year 6 grows at 4% and hands back most of its operating profit.

The terminal value

WACC 10 % SECOND-STAGE GROWTH 4 %

AT THE END OF YEAR 5	\$ MILLIONS
Year 6 free cash flow	27.39
Divided by the WACC less the growth rate	6.0%
Terminal value	456.50

VALUE PER SHARE **\$25.03**

Move the growth rate a half point and watch the denominator, not the numerator.

Discount it all back

WACC 10 % SECOND-STAGE GROWTH 4 %

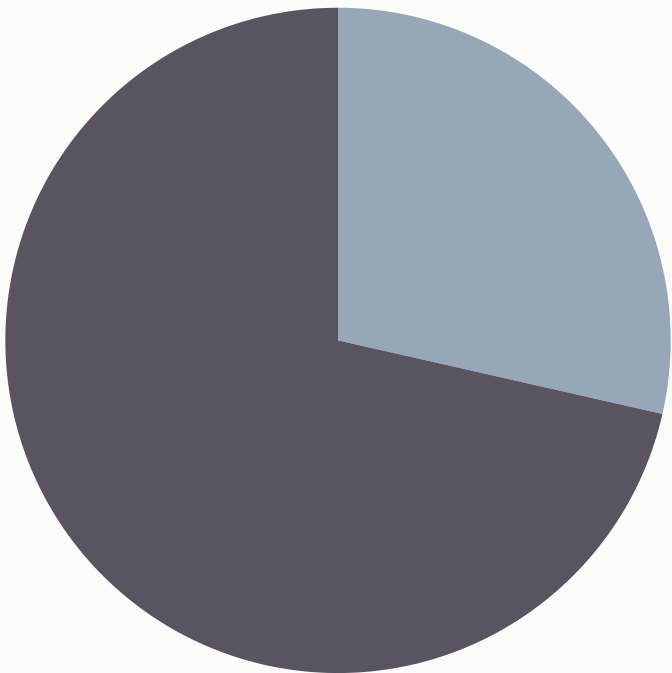
\$ MILLIONS	YR 1	YR 2	YR 3	YR 4	YR 5	TV AT YR 5
Amount	3.60	4.32	13.48	19.74	26.34	456.50
Discounted to today	3.27	3.57	10.13	13.48	16.35	283.45

ENTERPRISE VALUE **330.25** OF WHICH TERMINAL **85.8%**

The terminal value and the year 5 cash flow carry the same exponent.

Debt, cash, shares

DEBT 100 CASH 20 SHARES, MILLIONS 10



Enterprise value 330.25

Plus cash 20.00

Value of the two claims 350.25

■ Debt 100.00

■ Equity 250.25

Value per share \$25.03

The whole model

SALES 100 GROWTH 20 % 20 % 20 % 12 % 8 % THEN
4 %

MARGIN 20 % PP&E 45 % WORK CAP 15 % WACC 10 %

\$ MILLIONS	YR 1	YR 2	YR 3	YR 4	YR 5	YR 6
Sales	120.00	144.00	172.80	193.54	209.02	217.38
EBIT after tax (NOPAT)	18.00	21.60	25.92	29.03	31.35	32.61
Increase in invested capital	14.40	17.28	12.44	9.29	5.02	5.22
Free cash flow	3.60	4.32	13.48	19.74	26.34	27.39

ENTERPRISE VALUE **330.25** VALUE PER SHARE **\$25.03**

Move one input at a time

Second-stage growth

Take it from 4% to 5%, and then to 3%. How much of the value per share moves, and which term in the formula moved it?

Working capital

Take it from 15% of sales to 20%. In which years does the cash flow fall, and what has happened by year 6?

First-stage growth

Take years 1 to 3 from 20% to 25%. Does the value per share rise or fall, and what does the answer depend on?

The margin

Take it from 20% to 18%. Compare the change with what the growth rate did. Which assumption is the value more sensitive to?

Exercise: have AI build it

- 1 Upload this deck and ask for an Excel workbook containing the DCF model
- 2 Ask for an editable single-page HTML app that holds the model and explains it

Open the formulas in the workbook that comes back and check two things against this deck: which year's free cash flow sits in the numerator of the terminal value, and what exponent the terminal value is discounted at.

ANOTHER EXAMPLE

A second company

The forecast

Sales of 250 last year, growing 15%, 15%, 12%, 9%, 6%, and then 3.5% forever. Operating margin 14%, tax 25%.

The ratios

Net PP&E is 55% of next year's sales. Working capital is 10% of next year's sales.

The rest

WACC 9%. Debt 180, cash 35, twenty million shares. Figures in \$ millions.

Exercise: build it yourself

Build the spreadsheet model yourself, without AI help.

- 1 Free cash flow for years 1 through 6
- 2 Terminal value at the end of year 5, and the enterprise value
- 3 Equity value per share

The model should give \$17.50 a share, with the terminal value carrying 87% of the enterprise value.